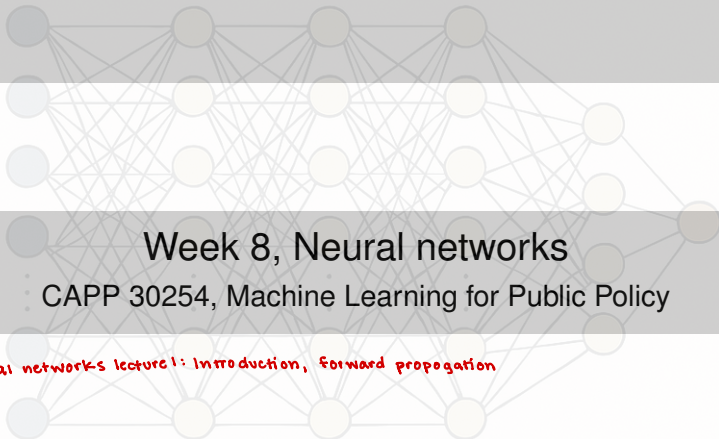




Week 8, Neural networks

CAPP 30254, Machine Learning for Public Policy

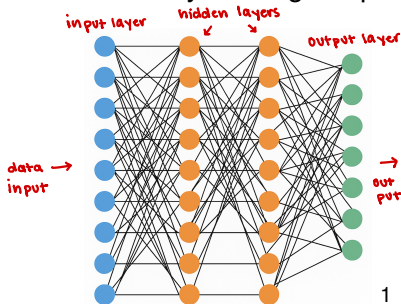
Neural networks lecture 1: Introduction, forward propagation

CAPP 30254
MACHINE LEARNING
FOR PUBLIC POLICY

Neural networks

(NN)

A *neural network* is a machine learning model that learns patterns from data by passing that data through layers of connected nodes. Each layer transforms the data a tiny bit so that the model will eventually make good predictions.



How big are NN?
of hidden layers

small	1-2
medium	3-10
large	10-100+
very large	100's, 1000's

deep learning
e.g., GPT

¹Generated by GPT-4, DALL-E 3 after the prompt: "Can you draw a picture of a complex neural network?"

Why use neural networks?

Neural networks are inspired by the brain's biological process. The brain is made of neurons and neurons receive signals, process them, and pass them out.

Now, neural networks power many modern AI systems. They can learn complex, nonlinear patterns way beyond what modern linear models can achieve.

They can:

- Translate languages *Natural language processing*
- Classify sentiment (e.g., positive vs. negative text)
- Recognize objects in images *Computer vision*
- Power self-driving car vision systems
- Play chess, Go, or video games

Neural networks in public policy

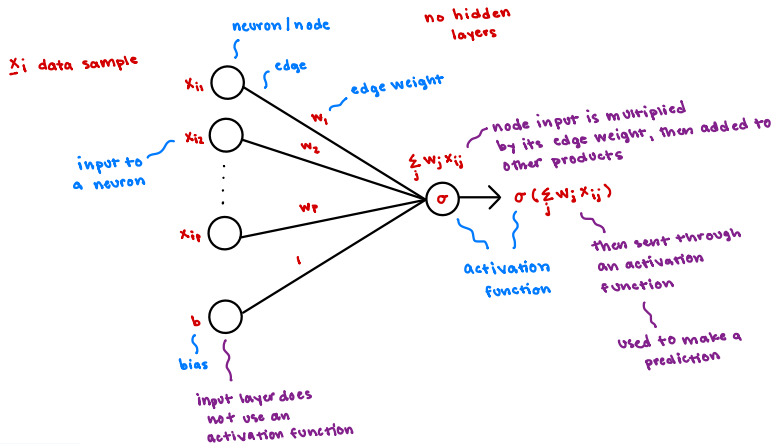
Neural networks are commonly used to find patterns, make predictions, and make decisions in public policy.

Some examples are:

- Detect abnormalities in X-rays, MRIs, and CT scans
- Optimize traffic flow using real-time data
- Model spread of diseases (e.g., COVID-19) or predict hospital demand
- Transcribe spoken language for language learners

A very simple neural network

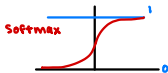
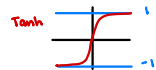
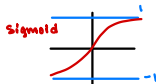
Neural networks can represent very simple models that make predictions. *input layer + output layer*



Activation functions

The output of a neuron is passed to an *activation function*, which transforms it. Some common activation functions are:

activation function choice depends on the task:



Name	Formula
ReLU (Rectified Linear Unit)	$\max(0, x)$
Sigmoid	$1/(1 + e^{-x})$
Tanh	$\tanh(x)$
Leaky ReLU	$\max(0.01x, x)$
Softmax	$e^{x_i} / \sum e^{x_j}$

common for hidden layers

binary classification output

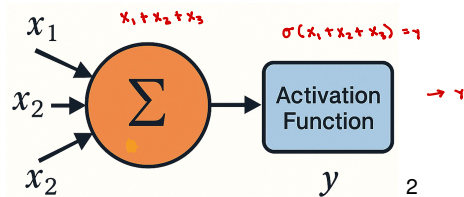
} if ReLU isn't working

multi-class classification output

Applying an activation function

Each hidden layer neuron usually applies an activation function to its input. This introduces nonlinearity, allowing the network to learn complex patterns.

Without any activation functions, the whole network would be equivalent to a single linear transformation.



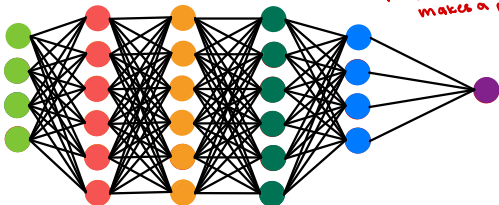
²Generated by GPT-4, DALL-E 3 after the prompt: "Can you show one node applying an activation function?"

Training a neural network

A neural network learns by passing data forward, measuring the prediction error using a loss function, and adjusting its internal weights.

Step-by-step, the process is:

- ① ■ Pass the inputs forward through the layers
 - ② ■ Apply activation functions
 - ③ ■ The network makes a prediction
- ① Ex. feature values for predicting voter turnout
- ② pass values through layers each layer does a little math (activation function) and passes forward
- ③ at the end (numbers reach the purple node) the network makes a prediction
- makes nonlinear patterns



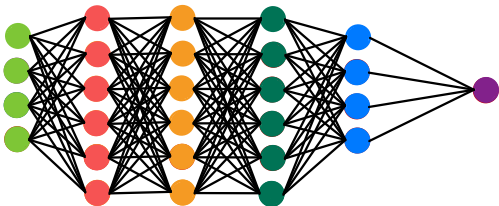
Training a neural network

NEXT TIME !!

A neural network learns by passing data forward, measuring the prediction error using a loss function, and adjusting its internal weights.

Step-by-step, the process is:

- Compare to the true label and compute the loss
- Use backpropagation and update the weights
- Repeat until the network learns

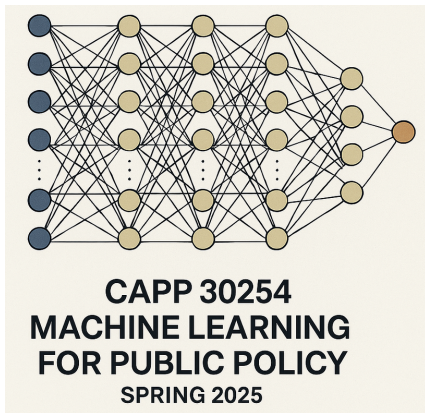


Backpropagation

After a forward pass, the error (loss) is calculated. The error is sent backward through the network to the input. An optimization algorithm like gradient descent is used to update the weights on the edges, which (hopefully) will help the prediction do better next time.

This process is called *backpropagation*.

Let's do a tiny example!



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³Generated by GPT-4, DALL-E 3 after the prompt: “Could you make a complex neural network with ‘CAPP 30254 Machine Learning for Public Policy, Spring 2025’”